

Workshop 12

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Answers

- The purpose of term weighting is that it assigns important scores to words, helping to identify relevant information, rank potential answers by relevance, and improve retrieval accuracy by highlighting critical terms in both the question and retrieved passages, filtering noise and focusing the search. BM 2.5 is a more advanced version of TF-IDF designed for better ranking in search engines and question-answering systems. In TF-IDF, if a word appears 100 times, it is roughly 10 times more important than if it appeared 10 times. This can lead to documents ranking too high just by repeating keywords. Whereas BM 25 uses a saturation function where the benefit of seeing a word again decline quickly. Once a word appears a few times, seeing it again barely increases the score. TF-IDF generally treats all documents as having equal length whereas BM 25 normalizes scores based on document length relative to the average. The last difference is tuning parameters where TF-IDF is primarily parameter-free, BM 25 includes k_1 (Controls how quickly term frequency saturates) and b (Controls how much document length influences the score).
- The main purpose of an inverted index is to enable fast full-text searches within large collections of documents or data. It functions by mapping individual terms (words or numbers) to the specific locations where they appear, allowing a system to identify relevant documents almost instantaneously without scanning every row or document in the entire collection.
- Modern IR systems have largely moved away from extensive stop word removal due to four key factors:
 - Semantic Integrity: Stop words are critical for preserving the intent of phrase queries (e.g., "flights to London") and specific titles. Removing them leads to significant precision loss and breaks disambiguation logic (e.g., "movies without Nicolas Cage").
 - Weighted Scoring (IDF): Advanced models like BM25 leverage Inverse Document Frequency (IDF) to automatically penalize frequent terms. This makes manual stop lists redundant, as the algorithm naturally discounts their impact on relevance.
 - Infrastructure & Optimization: Drastic reductions in storage costs and the adoption of index compression have mitigated the overhead of indexing common terms. Furthermore, impact-sorted indexes enable early termination, ensuring high-frequency terms don't bottleneck query performance.

- Query-Specific Handling: Global stop lists are a "blunt instrument." Modern systems prefer query rewriting and domain-specific logic, which offer more granular control than static, collection-wide word removal.
- The vocabulary mismatch problem occurs when query terms do not align with document terms despite semantic relevance (the "Human-Computer Interaction" naming problem). In IR, this is addressed through four primary technical strategies:
 - Morphological Normalization: Standardizes terms to a common root form via Stemming (algorithmic chopping) or Lemmatization (dictionary-based morphological analysis). This bridges the gap between variations like "running" and "runs."
 - Query Expansion (QE): Augments the user's query with additional terms via:
 - Lexical Resources: Utilizing thesauri (e.g., WordNet) for synonym injection.
 - Relevance Feedback: Utilizing terms from top-ranked documents to refine the query.
 - Document Expansion: Pre-processes documents at index time to improve coverage.
 - Doc2Query: Uses neural models to predict and append potential queries to a document.
 - Term Injection/Reweighting: Injects new terms or increases the frequency of existing keywords to optimize for the matching algorithm.
 - Semantic Matching: Moves from Syntactic Similarity (exact keyword matching) to latent representations:
 - Latent Semantic Indexing (LSI): Uses SVD to find hidden thematic relationships between terms (e.g., mapping "cellphone" to "mobile").
 - Controlled Vocabularies: Uses formal taxonomies/ontologies to map diverse user input to a standardized domain term.
- RAG is considered superior to standard prompting because it provides an LLM with a dynamic "cheat sheet" of external data rather than forcing it to rely only on its static training memory.

The primary advantages include:

- Factual Accuracy: It grounds the model's responses in verifiable, external documents, significantly reducing the risk of "hallucinations" (invented facts).
- Knowledge Freshness: RAG allows the model to access the most current information, such as live news or updated internal databases, bypassing the "knowledge cutoff" of the original training.
- Transparency: It provides source attribution, allowing users to see exactly which documents or paragraphs were used to generate an answer, which builds trust.
- Proprietary Context: It enables the model to securely reference private company data (e.g., manuals, emails, or technical docs) that was never part of its public training set.
- Cost Efficiency: Unlike fine-tuning, RAG updates the model's knowledge by simply updating an external database, avoiding the high computational costs of retraining.

- RAG process for question answering is typically divided into three primary stages: Data Preparation (Ingestion), Retrieval, and Generation.
 - Data Preparation (Ingestion)
 - This foundational stage occurs before a user ever asks a question. It prepares external knowledge for fast searching.
 - Retrieval
 - This stage begins when a user submits a query.
 - Generation
 - The final stage transforms the raw information into a natural language answer.

Refereneces

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